

CS 162–Computer Architecture

2026 Winter

Schedule

	Class
Time:	MWF, 9:00am – 9:50am
Room:	Room: WCH 143
Webpage:	http://eLearn.ucr.edu
Piazza (for discussions):	https://piazza.com/ucr/winter2026/cs162

Contact Information

Instructor: Bingyao Li		TA: Sixu Tan	
Office:	WCH 358	Office:	WCH 363
Email:	bingyaol@ucr.edu	Email:	stan073@ucr.edu
Office Hours:	Monday 10:00am-11:00am	Office Hours:	Thursday 3:00pm-4:00pm

Description

This course focuses on advanced (and some basic) concepts in modern computer architecture and the major subsystems of high-performance computers. Particular emphasis will be given to the exploration of parallelism at the instruction, data and thread levels. The course assumes students have basic knowledge of computer architecture from CS 161.

Textbooks

Computer Architecture: A Quantitative Approach; John Hennessy and David Patterson. Sixth Edition - Morgan & Kaufmann.

Reference book if you need review some basics of computer architecture: Computer Organization and Design - The Hardware/Software Interface By David Patterson and John Hennessy Fifth Edition - Morgan & Kaufmann.

Class Policies

Exams: There will be mid-term and final exams. The exams will be **close-note, in-class, in-person, synchronized exams**. If you cannot attend the exam, please contact the instructor as early as possible to discuss the reasons. If the reasons are acceptable, please coordinate with the instructor to schedule a conflict exam before the deadline. Cheating on exams will not be tolerated. Anyone caught cheating will be given a zero for the test and reported to the department following University procedures.

Project: There will be one programming course project. The logistics and details of the project will be given during lab section.

Homework: There will be homework that is meant to check your understanding of the course material. Totally, there will be 4 homework. Homework is individual work, so you are forbidden from sharing your solutions with your peers. Discussion is allowed and encouraged. Please list your discussion peers in your homework.

For all the above, no late submission is acceptable. These are meant to be your own work; anyone found to be cheating and copying will be disciplined in accordance with university policy. Cheating means (but is not limited to): using code/answers from previous terms, other universities, your friends, finding it on the Internet, getting help from unapproved forums, or outsourcing it. For project, we will be using Moss, a tool from Stanford for determining inappropriate collaboration.

Grading

Exams	40% (20%/each)
Project	40%
Homeworks	20% (5%/each)
Total	100%

Letter Grade	Percentage
A	> 92%
A-	> 89%
B+	> 86%
B	> 83%
B-	> 80%
C+	> 77%
C	> 73%
C-	> 70%

Academic Integrity

Here at UCR we are committed to upholding and promoting the values of the Tartan Soul: Integrity, Accountability, Excellence, and Respect. As a student in this class, it is your responsibility to act in accordance with these values by completing all assignments in the manner described, and by informing the instructor of suspected acts of academic misconduct by your peers. By doing so, you will not only affirm your own integrity, but also the integrity of the intellectual work of this University, and the degree which it represents. Should you choose to commit academic misconduct in this class, you will be held accountable according to the policies set forth by the University and will incur appropriate consequences both in this class and from Student Conduct and Academic Integrity Programs. For more information regarding university policy and its enforcement, please visit: <http://conduct.ucr.edu>.

Course Schedule

Daily topics are subject to change depending on our pace. They are there to assist you in reading the textbook so you can focus on those concepts prior to class.

*All assignment dates and due dates listed below are approximate, and subject to change.

	Date	Topic	Recommended Reading chapters	Assignments start	Assignments Due
Week 1	1/5	Course introduction and overview	Chapter 1		
	1/7	Performance			
	1/9	Review of instruction set	Appendix A		
Week 2	1/12	Review of pipelining	Appendix C		
	1/14	Branch/control hazards		HW1	
	1/16	Register renaming	Chapter 3		
Week 3	1/19	Martin Luther King's Day (No class)			
	1/21	Continue & Branch prediction		HW2	HW1
	1/23	Branch prediction			
Week 4	1/26	Tomasulo's algorithm			
	1/28	Tomasulo's algorithm speculation		Project	HW2
	1/30	Mid-term review			
Week 5	2/2	In-class mid-term exam 1			
	2/4	In-class mid-term exam 2			
	2/6	Review of memory hierarchy	Appendix B		
Week 6	2/9	Advanced cache optimizations	Chapter 2		
	2/11	Advanced translation optimizations		HW3	
	2/13	Main Memory			
Week 7	2/16	Presidents' Day (No class)			
	2/18	Thread Level Parallelism	Chapter 5		HW3
	2/20	Cache coherence			

Week 8	2/23	Directory-based coherence		HW4	
	2/25	Data level parallelism	Chapter 4		
	2/27	GPU architecture			
Week 9	3/2	GPU architecture (continued)			HW4
	3/4	Final review			
	3/6	In-class final exam 1			
Week 10	3/9	In-class final exam 2			Project
	3/11	Project			
	3/13				

UCR ACADEMIC POLICIES AND STUDENT SUPPORT

ITS Help and Student Technology Support

ITS Student Technology Services supports 9 student computer labs, including 7 public labs and 2 non-public labs, with approximately 293 public lab hours per week (with reduced hours during breaks and summer) available for academic use by all UCR students.

Whether you need a machine to work on your paper, need to use specialized software required by your class, assistance with your R'Mail or iLearn accounts, getting on the network, or any other services used by UCR students, contact BearHelp@ucr.edu.

Academic Misconduct

In addition to the course conduct policies outlined by your professor in the Course Syllabus in the online classroom, please review the Graduate Student Handbook.

It is the responsibility of each student to be familiar with the definitions, policies, and procedures concerning academic misconduct. Please revisit our Academic Integrity Policies and Procedures for more information. This site also defines misconduct, provides examples of prohibited conduct, and explains the sanctions available for those found guilty of misconduct.

Plagiarism

Plagiarism is the most common form of academic misconduct at UCR. It is the appropriation of another person's ideas, processes, results, or words without giving appropriate credit. This includes the copying of language, structure, or ideas of another and attributing (explicitly or implicitly) the work to one's own efforts. Plagiarism means using another's work without giving credit. Students who use ChatGPT and similar tools in improper ways, are violating the academic integrity rules of the University. Using ChatGPT as a study tool is acceptable, however, using ChatGPT to generate source code for assignments and submitting that would be in violation.

For more information about plagiarism, see Academic Integrity Policies and Procedures.

Reasonable Accommodation for Disabilities

UCR is committed to providing reasonable accommodation for all students with disabilities. Students with disabilities who require accommodations in this course should contact their professor as early in the semester as possible.

Students with disabilities must be registered with the Student Disability Resource Center prior to receiving accommodations in this course.

If you have a disability and you would like to make a request for reasonable accommodation, please see the Graduate Student Handbook or get in touch with the Student Disability Resource Center.

Adjustments for Pregnancy/Childbirth Related Issues

Should you need modifications or adjustments to your course requirements because of documented pregnancy-related or childbirth-related issues, please contact me as soon as possible to discuss your options. Generally, modifications will be made where medically necessary and similar in scope to accommodations based on temporary disability. Learn more about the rights of pregnant and parenting students by consulting the Office of Diversity, Equity, and Inclusion.

Title IX Resources

For any concerns regarding gender-based discrimination, sexual harassment, sexual misconduct, stalking, or intimate partner violence, the University offers a variety of resources, including advocates on-call 24/7, counseling services, mutual no contact orders, scheduling adjustments, and disciplinary sanctions against the perpetrator. Please see the Title IX website for more information. They can be reached at (951)827-7070. You can also file a report.

Student Needs

Student Health Services, Counseling & Psychological Services (CAPS), Residential Life, Dining, and R'Pantry are available to support students.

Religious Holidays

It is the policy of the University to excuse absences of students that result from religious observances and to provide for the rescheduling of examinations and additional required classwork that may fall on reli-

gious holidays without penalty. It is the responsibility of the student to make alternate arrangements with the instructor at least one week prior to the actual date of the religious holiday.

Copyright Policy

It is illegal to download, upload, reproduce, or distribute any copyrighted material, in any form and in any fashion, without permission from the copyright holder or his/her authorized agent. UCR expects all members of its community to comply fully with federal copyright laws.

Registration and Withdrawal

If you choose to withdraw from this course, you must complete the appropriate University form and turn the form in before the deadline. Deadlines are shown in the Academic Calendar, which is available from the Office of the Registrar.